

SANJAY KUMAR MAHANTA

Software QA Engineer — API, Web & Mobile Testing | AI-Assisted Testing

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India | Open to Remote (India & Worldwide)

PROFESSIONAL SUMMARY

QA Engineer / SDET with 3.6+ years of full-cycle testing experience — requirements review through release validation — across API, web, and mobile (Android/iOS) platforms. Skilled in functional, regression, smoke, and usability testing, with strong test strategy, test plan, and test case design. Hands-on experience using AI-assisted engineering tools (Claude Code, Cursor IDE, ChatGPT) to generate test scenarios, investigate defects, and speed up test design, while independently verifying AI-generated output before relying on it. Automates using Selenium (Java), Cucumber (BDD), Playwright (TypeScript), and a Pytest-based Python framework for UI/API and AI/LLM evaluation. Hands-on with Postman, REST Assured, SQL, CI/CD (GitHub Actions, Jenkins), and cloud environments (AWS, Azure).

TECHNICAL SKILLS

Requirements & Test Planning: Specification & Requirement Review, User Story Definition, Test Strategy Design, Test Plan Design, Test Case & Checklist Design, Test Design Techniques, Testing Basis Identification, Test Coverage Analysis, Effort Estimation Support, Root Cause Analysis

Manual & Functional Testing: Manual Test Case Design & Maintenance, Functional Testing, Regression Testing, Smoke Testing, Usability Testing, Release Testing, Exploratory Testing, Full-Cycle Testing (Requirements Review through Release Validation), Web & Mobile Testing (Android/iOS), Agile Ceremonies (Sprint Planning, Standups)

Automation Frameworks: Selenium WebDriver (Java) — primary, Cucumber (BDD), Page Object Model (POM), Pytest-based Python automation (RAGAS/LLM evaluation framework), Playwright (TypeScript), Appium — conceptual knowledge

Languages & Scripting: Java, Python, TypeScript

Test Frameworks: TestNG, JUnit

Version Control & CI/CD: Git, GitHub, GitHub Actions, Jenkins, Selenium Grid (Dockerized), automated test execution in pipelines

Operating Systems & Virtualization: Linux, working exposure to Windows/macOS via cross-platform web and mobile testing, Docker, AWS EC2 environment provisioning

Defect & Test Management: Jira, ClickUp, bug reporting, developer collaboration, bug-fix verification

API & Integration Testing: Postman, REST Assured, REST APIs, HTTP Methods, Status Codes, Swagger/OpenAPI

Mobile Testing: Android & iOS web/app testing, Chrome DevTools device emulation, LambdaTest cross-device validation, responsive/cross-platform coverage

Database Testing: MySQL

AI-Augmented Testing & AI Feature Validation: Claude Code, Cursor IDE, ChatGPT, n8n, RAGAS, LangChain, OpenAI API, Gemini API — AI-generated test cases/test data, AI-assisted self-healing test automation, validating AI-driven features (LLM/RAG output accuracy, relevancy, faithfulness)

Performance Testing: JMeter

Cloud & Containers: AWS (EC2, staging/production validation), Azure, Kubernetes, Terraform, Ansible

Security Awareness: OWASP Top 10 principles

Other: Figma review

WORK EXPERIENCE

QA Engineer — Interoffer (Independent Contractor)

Feb 2026 – Apr 2026

- Tested a real estate platform with a fully remote, distributed team — coordinated directly with developers to build module-level knowledge before each testing cycle, rather than testing blind against a spec.
- Reviewed Figma designs with the design team ahead of implementation to understand intended UI/UX behavior, then validated the built module against those designs during testing to catch visual and interaction mismatches early.
- Automated E2E UI and API validation using Selenium (Java) with Page Object Model (POM) design, alongside browser DevTools for debugging failed runs.
- Performed cross-browser and mobile web testing (Android/iOS) using Chrome DevTools device emulation to validate responsive, functional, and usability behavior across screen sizes and devices.
- Ran full-cycle functional testing across both backend and frontend of each module — from requirements review through release validation — manually validating REST API endpoints and status codes via Postman, and cross-checking SQL queries for client-server data integrity.
- Logged and tracked defects with clear reproduction steps and evidence in ClickUp, keeping a single source of truth on bug status across the remote team.

Automation Tester — GreenITCo Technologies

May 2025 – Sep 2025

- Owned end-to-end QA directly under the Product Manager for an ITM (IT Asset Management) tool, working closely with the product side to understand feature intent before test design — not just executing handed-down cases.
- Built the team's Selenium (Java) automation framework from scratch — first ground-up framework of its kind at the company — using TestNG for test orchestration and Page Object Model (POM) for maintainability, architected to run on a Dockerized Selenium Grid, with code pushed to GitHub and triggered automatically via Jenkins for continuous regression coverage.

- Extended the framework's reach when cross-browser and mobile testing needs came up mid-project: integrated the BrowserStack SDK to run test cases on LambdaTest across browsers, Android, and iOS devices, turning a one-off manual ask into a fully automated CI/CD-driven cross-platform pipeline.
- Validated REST APIs with Postman, covering authentication flows and edge-case status codes ahead of releases, alongside functional, regression, smoke, and usability checks each release cycle.
- Reviewed product and support documentation for technical accuracy and English/grammatical correctness, helping developers catch content issues before release.
- Identified a recurring legal-risk issue where flagged competitor terms kept slipping into front-end copy; instead of manually removing them each time, self-hosted an n8n workflow on Docker (avoiding paid SaaS costs) that scanned all documents, extracted findings into structured docs, and auto-emailed a report via Gmail — eliminated repeat manual scanning and made the fix future-proof, since resolving a new term going forward only needs a prompt update, not a re-scan.

Software Test Engineer — Amedea Infotech

Jan 2022 – Jan 2025

- Owned end-to-end manual QA cycle for assigned modules on a large-scale E-Commerce platform — translated feature requirements from leads into detailed test strategies, test scenarios, and test cases in Excel, walked leads through the coverage for sign-off before execution began, ensuring nothing shipped untested.
- Ran exploratory, functional, regression, smoke, and usability (black-box) testing on approved scenarios, applying test design techniques to identify testing basis and proactively pulling domain context and edge-case clarification from developers to catch gaps that scripted test cases alone would have missed.
- Logged and tracked defects with clear reproduction steps in the internal bug-tracking tool, and compiled Excel-based test summary reports for leadership when a formal report was needed — kept defect visibility high across both fast-moving sprints and release cycles.
- Extended coverage into API testing using Postman when releases needed it — validated REST endpoints, HTTP methods, and status codes, and logged defects with request/response evidence to speed up developer triage.
- Transitioned into automation as the team's needs grew: wrote Selenium WebDriver (Java) test scripts using TestNG and Page Object Model (POM) for the highest-value regression scenarios and ran them through a Dockerized Selenium Grid hosted on AWS, cutting down repetitive manual regression work over 36 months on the platform.
- Load tested critical flows up to 500 concurrent users with JMeter, helping resolve 3 critical performance bottlenecks.
- Worked with the team to provision AWS staging and production environments using Terraform and Ansible, so infrastructure setup for new test cycles stopped being a manual bottleneck.

ACHIEVEMENTS

AI-Powered Self-Healing Test Framework

- Built an intelligent framework using Gemini API to auto-detect and repair broken DOM selectors at runtime — reducing script maintenance time by 30–40% and cutting down on flaky-test overhead.

Legal Compliance Document Review Pipeline (n8n + Docker + Gemini API) — GreenITCo

- Built and deployed a fully automated document review pipeline on Docker, integrating n8n as the orchestration layer with Gemini as the LLM engine — replacing a manual 1-hour legal compliance task with a 20-second automated workflow (99% time reduction).
- Ingested legal/compliance documents, extracted competitor keywords flagged as compliance risks, logged structured results to Google Sheets, and delivered summarized findings via Gmail — zero manual intervention required, including validation of AI-generated flagging accuracy.

PROJECTS

- E-Commerce Platform — Amedea Infotech: Functional, regression, smoke, usability, performance, and API testing across a large-scale online retail platform over 36 months.
- ITM (IT Asset Management) Tool — GreenITCo Technologies: End-to-end QA under the Product Manager, with automation framework built from scratch for regression coverage across web and mobile.
- Real Estate Platform — Interoffer: End-to-end manual and automated testing across backend and frontend modules, including mobile web validation, for a remote real estate product.

CERTIFICATIONS

- AI Testing with DeepEval — Rahul Shetty Academy (2026): Hands-on evaluation of AI Agents, Chatbots, and RAG applications using DeepEval — covering evaluation metrics (Answer Relevancy, Faithfulness, G-Eval), Golden Dataset creation, trace-based and component-level testing, synthetic test data generation, and AI safety testing.
- Foundations of Prompt Engineering — AWS

EDUCATION

- Master of Computer Applications (MCA) — Savitribai Phule Pune University (SPPU), 2019, 65%
- B.Sc. Statistics — University of Pune (SPPU), 2013